Type MS · State the variables to control in an investigation

Answer 1 of 21 · Question 2 part a.ii · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

2 (a)	(i)	• a tube containing inhibitor and substrate; • but no enzyme;		2
	(ii)	any three from • concentration of enzyme; • concentration of inhibitor; • temperature; • pH; • same filter in colorimeter;		3
(b)	(i)	axes labelled with units; axes correct way round; plots;		3
	(ii)	suitable curve;		1
	(iii)	similar shaped curve; reaching maximum sooner;		2
	(c)	same maximum never reached; despite increase in substrate concentration;		2

Total 13 marks

Type MS · State the variables to control in an investigation

Answer 1 of 21 · Question 2 part a.ii · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 2 of 21 · Question 2 part c.ii · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c.ii in this mark scheme extract

2(a)	Any two from: carbohydrates/proteins/lipids/mineral salts (or named example)/fibre;;	Ignore named examples (1 mark for each two correct)	**(2)**

Question Number	Answer	Additional guidance	Mark
2(b)	Any two from: • lack of it causes scurvy; • (lack of vitamin C causes) damage to membranes/bleeding gums; • makes connective tissue/maintains/strengthens cell linings/gums/sticks/joins cells/tissues together;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)(i)	Any five from: • Add fresh juice/long-life juice (to a test tube); • add DCPIP drop by drop/slowly; • until blue colour disappears/solution turns colourless; • count number of drops/measure volume of DCPIP added; • repeat (to obtain mean); • compare volumes of DCPIP added;		**(5)**

Question Number	Answer	Additional guidance	Mark
2(c)(ii)	Any two from: • volume of juice; • temperature; • concentration DCPIP;	Allow amount for mp1. Ignore concentration	**(2)**

Total 11 marks

Type MS · State the variables to control in an investigation

Answer 2 of 21 · Question 2 part c.ii · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c.ii in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · State the variables to control in an investigation

Answer 3 of 21 · Question 9 part b.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
9	(a) (i)	• reference to platelets; • fibrinogen converted to fibrin; • (fibrin) mesh/crosslinks/fibres formed;		3
	(ii)	• reduced blood flow to heart/cardiac muscle / tissue/cells; • less oxygen/glucose delivered; • less (aerobic) respiration/energy released (by cardiac tissue);		3
	(b)(i)	size of blood clot/volume of water /aspirin solution used/temperature/shape of blood clot/width of straw;	Allow valid alternatives	1
	(ii)	use water without aspirin dissolved;		1
	(iii)	• the more tablets dissolved the more solution collected; • the more tablets dissolved the quicker the solution is collected; • aspirin dissolves/acts on blood clot; • size of blood clot reduced more quickly with more tablets/blood clot dissolves/breaks down faster with more tablets;		Max 3
	(c)	Any four from: • aspirin binds to enzyme/aspirin competes (with substrate) for active site; • active site of enzyme changed/blocked; • less/no substrate binds (to enzyme); • reaction involving release of chemicals reduced/stopped; • platelets do not stick together;		Max 4

Total question 9 = 15 marks

Type MS · State the variables to control in an investigation

Answer 3 of 21 · Question 9 part b.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · State the variables to control in an investigation

Answer 4 of 21 · Question 2 part c · 4 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

2 (a) (i)	better transfer of heat to water (1)		1
(ii)	any two from • safety goggles/glasses (1) • water may spit (1) • use test tube holder (1) • to avoid burning hands (1)	reason linked to precaution	2
(b) (i)	• 28 x 12 x 4.2 (1) • 1411/1411.2J (1) • 1400J (1)	error carried forward full marks for correct answer	3
(ii)	• some energy lost to air/absorbed by tube/needle (1) • some converted to light (1) • food not fully burnt;		3
(c)	• use same mass of food (1) • same volume of water (1) • use same thin tube (1) • burn food same distance from tube (1)		4

Total 13 marks

Type MS · State the variables to control in an investigation

Answer 4 of 21 · Question 2 part c · 4 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 5 of 21 · Question 7 part b.i · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Question number					
7	(a)	(i)	• allows protein to stay longer in stomach/gut (1) • gives time for full digestion (1)		2
		(ii)	• diet largely/entirely milk when young (1) • changes to solids when older (1)		2
		(iii)	• biuret test (1) • add reagent to solid (1) • blue to lilac/pale purple (1)		3
	(b)	(i)	• temperature (1) • volume of milk (1) • volume/concentration of chymosin (1)		3
		(ii)	$\frac{238 + 232 + 241 + 229}{4}$ (1) = 235 (1)		2
		(iii)	not included when calculating mean (1)		1
		(iv)	bubbles of different sizes/volume of CO_2 variable (1)		1

Total for this question = 14 marks

Type MS · State the variables to control in an investigation

Answer 5 of 21 · Question 7 part b.i · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 6 of 21 · Question 4 part a.ii · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

4 (a)	(i)	• axes labelled (1) • suitable scale (1) • axes correct way round (1) • correct plots (1) • lines labelled/key given (1)		5
	(ii)	any two from • volume of water (1) • mass of carbohydrate/aspirin (1) • temperature (1) • time interval between each reading (1)	Allow type of aspirin Allow amount as an alternative to mass Ignore concentration	2
	(iii)	pH (1)		1
(b)		• (aspirin) dissolves/reduces pH more quickly (1) • (when combined) with sucrose (1) • so will act more quickly/relieves pain faster (1)		3

Total 11 marks

Type MS · State the variables to control in an investigation

Answer 6 of 21 · Question 4 part a.ii · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 7 of 21 · Question 7 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
7	(a) (i)	• safety glasses/goggles; • care using sharp needle; • care to avoid burns; •		3
	(ii)	• mass of food; • volume of water; • distance of burning food from tube;	reject amount	3
	(iii)	any five from • not all food burns to release energy; • not all heat transferred to water; • some lost to atmosphere; • some heat lost to needle/glass tube; • uneven distribution of temperature in water; • because lack of stirring;		5
	(b)	any four from • use of oxygen; • allows complete combustion; • enclosed/lid so no heat escapes; • heat transfer coil so all heat is used to increase temp of water/transferred to water; • stirrer to distribute heat;		4

Total 15 marks

Type MS · State the variables to control in an investigation

Answer 7 of 21 · Question 7 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 8 of 21 · Question 7 part a.i · 3 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
7 (a)	(i)	• temperature of incubation; • volume of milk; • composition of nutrients in agar/type of agar;		3
	(ii)	size of bacterial growth;		1
(b)	(i)	$\frac{7.1}{5} = 1.42$; $\frac{36.5}{5} = 7.3$;	One mark for each correct answer	2
	(ii)	• more bacteria in B; • at start of investigation; • because extra added to tube; • some bacteria in A/milk which grew;	Allow more growth in B/growth increases more quickly in B Allow reference to contamination e.g.from air	4
(c)		• transfer loop sterilised/heated in flame; • transfer performed quickly;	Allow reference to sterilised in context / sterile technique / sterilise equipment / named equipment	2
(d)		• repeat investigation; • using other organisms;		2

Total 14 marks

Type MS · State the variables to control in an investigation

Answer 8 of 21 · Question 7 part a.i · 3 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 9 of 21 · Question 3 part a.ii · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

3 (a) (i)	Any five from the following: • place water in the test tube; • take the (starting) temperature of the water; • attach the test tube to a clamp stand; • measure the mass of a piece of food/weigh food; • place food on the mounted needle (under the test tube); • burn the food; (using a Bunsen burner) • take the (final) temperature of the water;		5
(ii)	Any two from the following: • volume/mass of water; • distance of the food from the test tube; • mass of food;	ignore amount ignore amount/size	2
(iii)	Any two from the following: • hold the burning food at arm's length; • reference to sharpness of needle; • goggles; • reference to food allergies/don't taste food;	any ref to being burnt/hot water	2
(b) (i)	Any two from the following: • males need more energy than females; • the amount of energy required decreases with age; • the amount of energy needed by males decreases more (overall) than females;		2
(ii)	**D** - Males use more muscle mass than females;		1
(iii)	• (2629 – 2103) = 526; • 526 ÷ 2103; • (x 100) = 25 (%);	if divide by 2629 and give 20% then two marks Total 15 marks	3

Type MS · State the variables to control in an investigation

Answer 9 of 21 · Question 3 part a.ii · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 10 of 21 · Question 1 part e · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part e in this mark scheme extract

1 (a)	• apparatus in correct order with T piece connected to both flasks; • long tubes in indicator; • short tubes above indicator;		3
(b)	limewater;	Allow sodium hydrogen carbonate	1
(c)	any four from: • breathe through mouth piece; • in and out; • 10 times/max of one minute; • compare contents of flasks; • exhaled flask should go cloudy if more carbon dioxide exhaled;		4
(d)	sterilise mouth piece/breath gently;		1
(e)	same volume of indicator in each flask;		1

Total 10 marks

Type MS · State the variables to control in an investigation

Answer 10 of 21 · Question 1 part e · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part e in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 11 of 21 · Question 6 part a.ii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
6	(a)	(i)	18-12; 6;	accept 17.8/17.9	2
		(ii)	Two from: • volume of milk; • volume of the starter culture/live yoghurt; • type of starter culture • type of milk; • type of live yoghurt; • mixing;	reject 'amount'	2
		(iii)	• repeat the investigation at more temperatures; • between 20°C and 60°C;		2
		(iv)	• heated to kill (unwanted) bacteria in the milk; • cooled so that the bacteria in the starter culture are not killed;		2
	(b)		Any two from: • pH 4.4 inhibits the bacteria/bacterial growth; • denatures enzymes; • changes the shape of the active sites • lactose cannot bind to the enzyme/no enzyme-substrate complexes formed;		2
	(c)	(i)	Any two from: • the beads can be re-used; • product doesn't need to be treated to remove the enzyme; • large surface area to increase the rate of reaction; • enzyme more temperature stable;		2
		(ii)	• long/villi/microvilli to increase surface area; • concentration gradient maintained with blood flow; • short diffusion distance;		3

Total 15 marks

Type MS · State the variables to control in an investigation

Answer 11 of 21 · Question 6 part a.ii · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 12 of 21 · Question 3 part b.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

3 (a)	(i)	surface area = 24 (mm^2);
		volume = 8 (mm^3);
		3:1;
			Full marks for correct final answer	3

	(ii)	C 92;		1
	(iii)	movement from a high concentration to a low concentration;		1
	(iv)	• reduces the time taken to become colourless/rate of diffusion/reaction is faster; • the diffusion distance is shorter/greater area over which diffusion/collisions can occur;		2
(b)	(i)	• use acid heated to different temperatures/idea of different temperatures; • compare/measure the time taken to become colourless/compare rate of diffusion;		2
	(ii)	Any two from: • the side length of the cube/size of the cube/surface area/volume of cube; • acid concentration; • concentration of alkali/indicator in the cube;	accept volume of acid Allow volume of indicator	2

Total 11 marks

Type MS · State the variables to control in an investigation

Answer 12 of 21 · Question 3 part b.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 13 of 21 · Question 2 part b.iv · 1 mark

2023 JUN · Paper 02 (regional R) · find part b.iv in this mark scheme extract

2 (a)	(i)	in the following order: • proteases; • protein Q; • amino acids;	Reject protein P	3
	(ii)	stomach/ small intestine/ pancreas/ duodenum/ ileum;		1
	(iii)	DNA / mRNA / messenger RNA;		1
(b)	(i)	• as pH increases activity increases; • to pH2/peaks at pH2/to optimum; • decreases from pH2;	Allow increases and then decreases for one mark	3 exp
	(ii)	Any two from: • enzyme is denatured (at pH2); • change in shape of active sites; • no enzyme-substrate complexes formed/no binding with substrate;		2 exp
	(iii)	• repeat; • calculate a mean/average/remove anomalous result		2 grad
	(iv)	Any one from: • temperature; • substrate concentration/volume; • enzyme concentration/volume;		1 grad

Total for Question 2 = 13 Marks

Type MS · State the variables to control in an investigation

Answer 14 of 21 · Question 7 part a.i · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

7 (a)	(i)	• volume/concentration of enzyme/lipase; • volume/concentration of indicator; • volume/same type of olive oil; • volume/mass of bile saltes	Ignore amount for all mps	3
	(ii)	40°C;		1
	(iii)	no values given between 30°C and 40°C; no values given between 40°C and 50°C; optimum could be between temperature intervals given/below/above 40ºC/50ºC;		3
(b)	(i)	• tube 3 = 11.5 hours and tube 8 = 24 hours; • 24 - 11.5 = 12.5 hours;	Ecf incorrect values in a subtraction calculation with a correct answer 1 mark	2
	(ii)	• lipase digests/breaks down olive oil; • produces fatty acids; • these acids lower the pH;		3
(c)		• no bile salts so no emulsification of oil/oil not broken down to droplets; • less surface area (for lipase to work on); • lipase only working very slowly/not very active/takes longer to digest oil;		3

Total 15 marks

Type MS · State the variables to control in an investigation

Answer 14 of 21 · Question 7 part a.i · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · State the variables to control in an investigation

Answer 15 of 21 · Question 2 part d · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

Question Number					
2	(a)	(i)	• axes labels; • plots; • scale;	axes correct way round use aty least half of the graph paper for the plotted results ignore extrapolation of line	3
		(ii)	line of best fit;		1
		(iii)	As time increases pH decreases;	ECF from an incorrectly drawn graph	1
	(b)		Any three from: • lipase; • digests/breaks down fat (in milk); • to fatty acids; • which lowers pH/increases acidity;		3
	(c)		time;	allow mins/minutes	1
	(d)		Any two from: • temperature; • volume/concentration of enzyme/substrate (solution)/volume of milk; • type of milk;	ignore amount	2
	(e)		repeat the experiment		1

Total Marks 12

Type MS · State the variables to control in an investigation

Answer 15 of 21 · Question 2 part d · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 16 of 21 · Question 4 part b.iii · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

4	(a)	(i)	• linear scale/plotted on half of grid; • correct way round; • axes labelled with units; • all points plotted correctly;	mp4- half-square tolerance	4
		(ii)	draw curve of best fit;	Ignore beyond the 2 extremes when line is drawn. 2 straight lines=0 Slight curve at 4 is enough for 1 mark	1
	(b)	(i)	• pepsin digests/breaks down protein; • (to)amino acid/peptides; • (which is)soluble/dissolves; • black silver salts removed /fall away/no longer present;	Mp1- accept hydrolyse MP4- accept silver salts broken down	3
		(ii)	• 1//2/1 to 2; • optimum pH for pepsin;		2
		(iii)	• temperature; • area of film/size/length/width/thickness/; • volume of (pepsin) solution; • type of photographic film/amount of protein in film; • pH;	Mp3-ignore amount	3
	(c)		• adding more enzyme/pepsin has no effect on reaction/ maximum rate of reaction at 4%; • all protein/substrate digested/ substrate is a limiting factor/not enough protein to digest; • enzyme no longer a limiting factor/ active sites empty;	MP1- accept Vmax not enough/no more substrate/protein to bind to active sites= 2 marks	2
					Total = 15

Type MS · State the variables to control in an investigation

Answer 16 of 21 · Question 4 part b.iii · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 17 of 21 · Question 4 part c.ii · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

4 (a) (i)	<table><tr><th>Conc (sugar) in au</th><th>Percentage change in mass</th></tr><tr><td>0.00</td><td>8.0</td></tr><tr><td>0.25</td><td>4.0</td></tr><tr><td>0.50</td><td>-1.0</td></tr><tr><td>1.00</td><td>-7.0</td></tr><tr><td>1.50</td><td>-10.0</td></tr></table> • two row/column table; • concentration (sugar) in au; • percentage/% change in mass; • 0.0 to 1.50/1.50-0.00;	Ignore units in the body of the table Accept AU for MP2 Ignore % symbol for MP3 mp3 unit not required for percentage in mass	4
(ii)	0.50 (au);		1
(iii)	$\frac{104 \times 7}{100}$; = 7.28 g; =7.3;	Allow 4/100 for 1 mark or 0.28	3
(b)	• change in mass is -1% to +4% • 5% range in mass • between 0.25 and 0.5 au; • accept any answer between 0.375- 0.5; ***an answer of o.45 scores =4 marks***	MP 1-4 are separate MPs unless the answer is 0.45	4
(c) (i)	• dry the cylinder/ remove water/liquid; • affects mass/wrong mass/heavier/inaccurate mass/eq;	Accept weight	2
(ii)	any two from • volume of (sugar) solution; • temperature; • surface area of cylinders; • same type of potato;	Ignore amount accept volume of liquid	2 **Total =16**

Type MS · State the variables to control in an investigation

Answer 17 of 21 · Question 4 part c.ii · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 18 of 21 · Question 2 part c · 1 mark

2020 JAN · Paper 02 · find part c in this mark scheme extract

Number				
2	(a) (i)	column headings(1) units in column headings(1) values correctly written in columns(1) Type of activity / exercise: at rest; walking; jogging; running Pulse/heart rate in beats per minute: 78; 95; 138; 146	Allow bpm for beats per minute	3
	(b)	Any four from: • take pulse rate at rest(1) • using monitor/from wrist/neck(1) • walk (for one minute)(1) • take pulse rate after walking(1) • repeat for jogging and running(1) • rest between each exercise(1)		4
	(c)	exercise for the same amount of time/same terrain/environmental conditions/wear same clothing		1
	(d)	breathing/metabolism/respiration/blood pressure/sweating		1

Total for Question 2 = 9 marks

Type MS · State the variables to control in an investigation

Answer 19 of 21 · Question 1 part b.i · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

number					
1	(a)	(i)	• first convoluted tubule; • loop of Henlé; • second convoluted tubule;	diagram only, no labels required	3
		(ii)	44mm; 44 ÷ 650; = 0.068mm;	ecf from mp1 max 2 marks Allow 0.0676	3
		(iii)	• protein; • red/white blood cells/platelets;		2
		(iv)	glucose;		1
	(b)	(i)	Any 2 from: • same person; • same activity levels; • same volume of water; • same food eaten;		2
		(ii)	710 ÷ 4; = 178 cm^3 per hour;	Allow 1 mark if division by 4 is shown Allow 177.5	2
		(iii)	• less urine production/lower urine loss on day two; • more ADH secreted on second day; • more water reabsorbed; • more water lost by sweating;	Allow reverse argument for all marking points	3

Total 16 marks

Type MS · State the variables to control in an investigation

Answer 19 of 21 · Question 1 part b.i · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 20 of 21 · Question 5 part a.ii · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
5	(a) (i)	any two from • to allow the heart rate to return to normal; • allow respiration/breathing rate/body temperature to return to normal; • to allow sweat to evaporate;		2
	(ii)	any three from • same time for exercise; • same type/intensity of exercise; • wear same clothing for each test; • measure sweat from same area;		3
	(b) (i)	• axes labelled with units with independent variable on x-axis; • suitable scale; • both sets of data plotted correctly;; • suitable lines; • lines labelled;		6
	(ii)	any five from • exercise increases muscle activity; • more energy/ATP required; • increased respiration; • heat produced as by product; • lost from body; • by increased sweating; • increase in room temperature increases rate of sweating;		5

Total 16 marks

Type MS · State the variables to control in an investigation

Answer 20 of 21 · Question 5 part a.ii · 3 marks

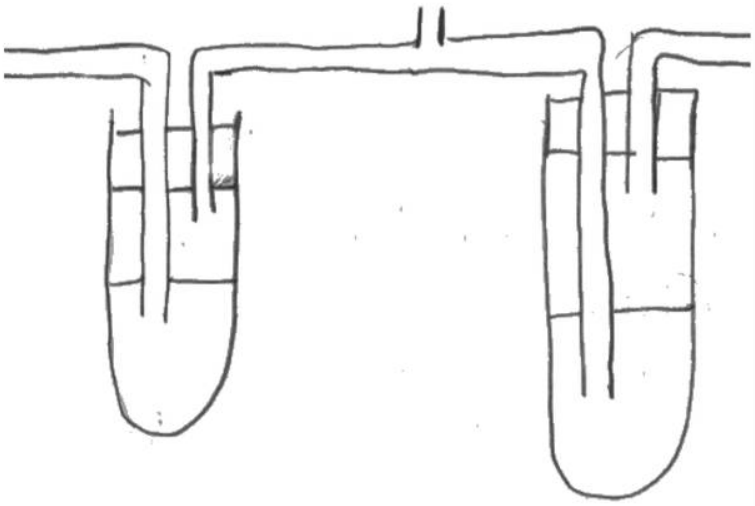
2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State the variables to control in an investigation

Answer 21 of 21 · Question 4 part b.i · 1 mark

2024 JUN · Paper 01 · find part b.i in this mark scheme extract

4	(a)	(i)	• apparatus correctly set up; • long tube in indicator; • short tube above indicator; 		(3)
		(ii)	Inhale and exhale gently/sterilise/clean mouthpiece/don't inhale liquid/care in assembling apparatus to avoid breaking glass;		(1)
	(b)	(i)	exercise/rest for the same length of time/at the same pace for each repeat/same student;		(1)
		(ii)	any four from • breathing rate increases (with vigorous exercise); • more oxygen inhaled/supplied; • for aerobic respiration; • more energy released/more ATP; • breakdown of lactic acid; • produced by anaerobic respiration;	ignore ref CO_2	(4)

Total for Question 4 = 9 marks